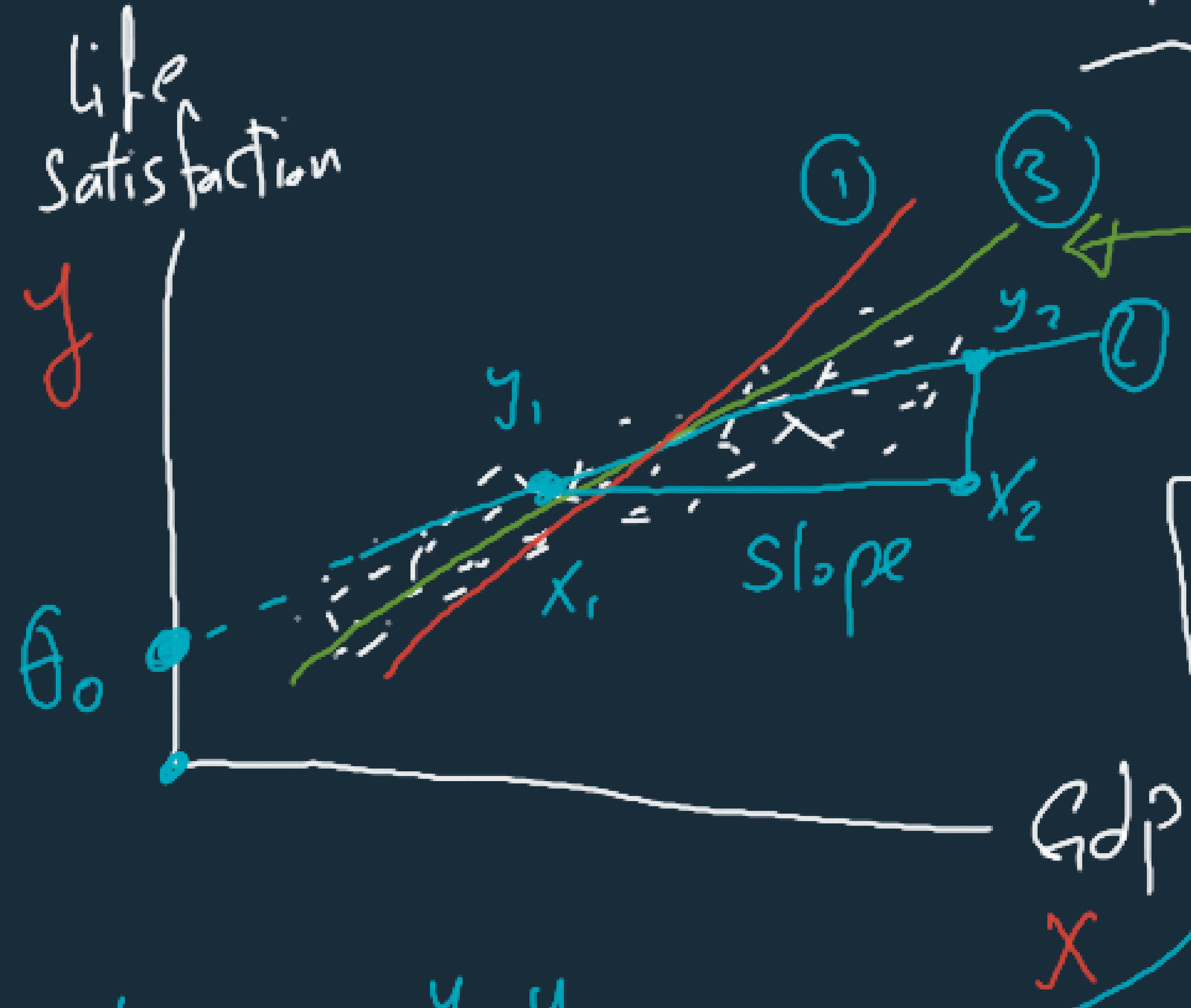
Linear Regression



Best fit line

$$\boxed{y = \theta_0 + \theta_1 x}$$

y → prediction [Dependant variable]

θ_0 → intercept (Bias)

θ_1 → Slope (Coeff)

x → feature [Independant variable]

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

features $\underbrace{x_1 \; x_2 \; x_3}$ label y → ML Model

y

- Continuous → $\boxed{\text{Regression}}$ → House price
- Categories → Classification → Spam filter

prediction ← $\hat{y} = \theta_0 + \theta_1 x$

??

⇓

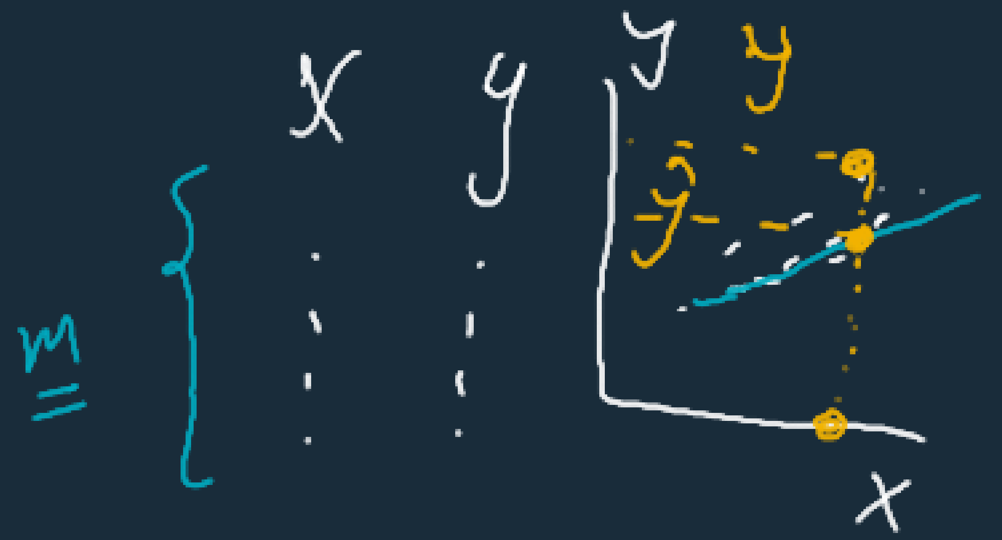
Best prediction

⇓

Min. Error → y (Actual), $\hat{y}$ (prediction) } Cost function

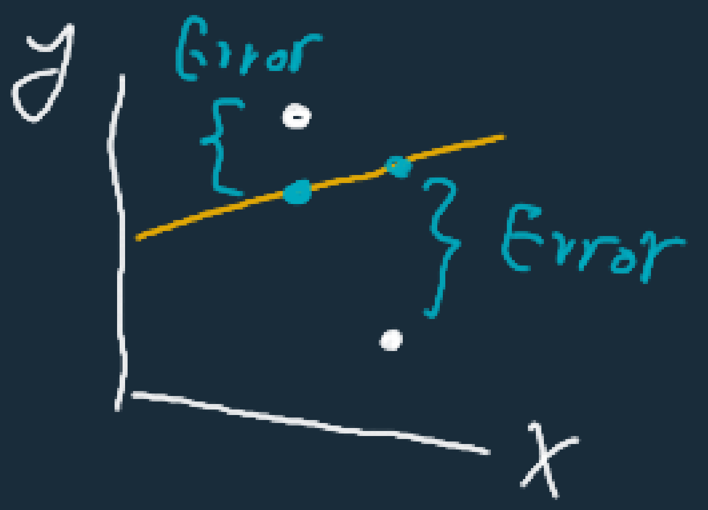
↳ Total Error (all points)

y $\hat{y}$



Cost function $= \frac{1}{m} \sum_{i=1}^{m} \left(y^{(i)} - \hat{y}^{(i)} \right)^2$

MSE



$$\hat{y} = \theta_0 + \theta_1 x \Rightarrow \text{Cost function } J = \frac{1}{m}\sum_{i=1}^{m}\left(y^{(i)} - \hat{y}^{(i)}\right)^2$$

Min

$$\boxed{\underset{\text{Min}}{J(\theta_0, \theta_1)} = \frac{1}{m}\sum_{i=1}^{m}\left(y^{(i)} - \theta_0 - \theta_1 x^{(i)}\right)^2}$$

Analytic

Closed form solution

Ordinary Least Squares (OLS)

$$\left.\begin{aligned} \frac{\partial J}{\partial \theta_0} &= 0 \\ \frac{\partial J}{\partial \theta_1} &= 0 \end{aligned}\right\} \quad \boxed{\begin{aligned} \theta_0 &= \checkmark \\ \theta_1 &= \checkmark \end{aligned}} \Rightarrow$$

Min Error

Best fit line

θ_0, θ_1 (Best fit line)

OLS

$\theta_0 = ✓$

$\theta_1 = ✓$

$$\boxed{\hat{y} = \theta^T X}$$

$$\boxed{\hat{\theta} = (X^T X)^{-1} X^T y}$$

→ Complexity

Normal Equation

θ_0 ✓

θ_1 ✓

vectorized

$X \rightarrow$ Matrix

$\hat{y} \rightarrow$ Matrix (vector)

$\theta \rightarrow$ Matrix $[\theta_0, \theta_1]$

Gradient Descent

(Iterative)

$J(\theta_0, \theta_1)$

→ Random values θ_0, θ_1

→ $\hat{y}(\theta_0, \theta_1)$

→ Error → gradient

↓

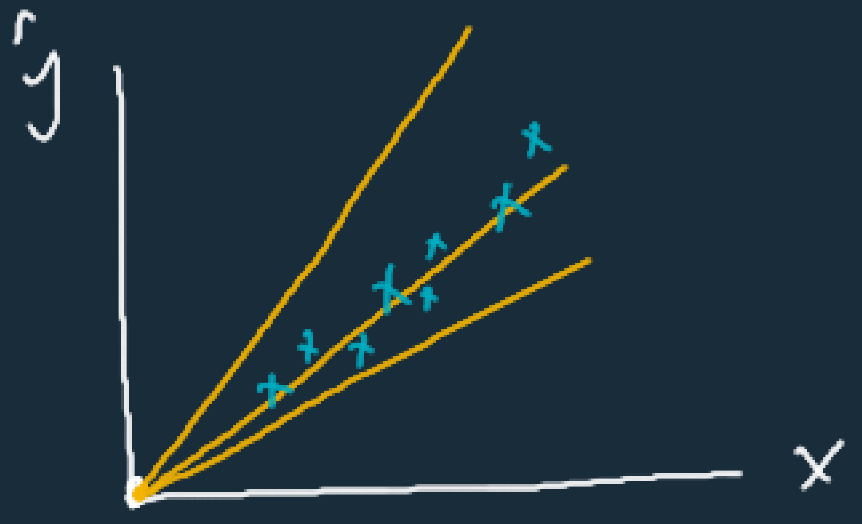
direction

$$\theta_{new} = \theta_{old} - \alpha \frac{\partial J}{\partial \theta}$$

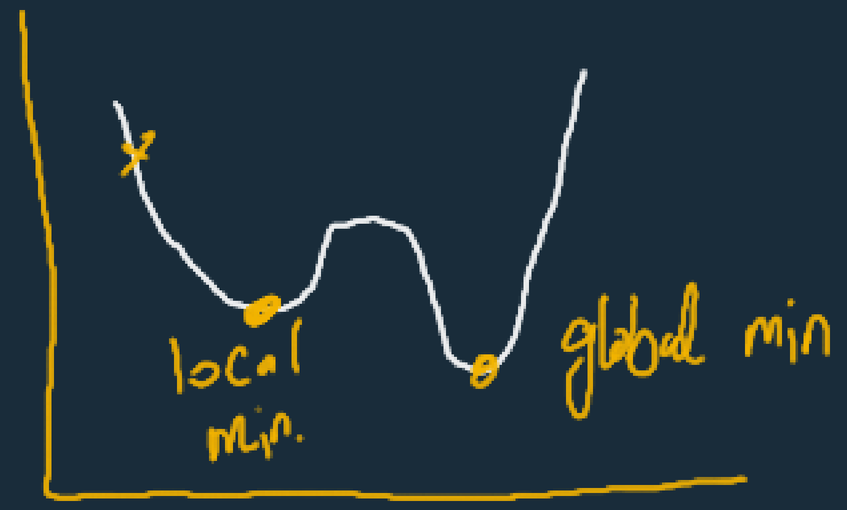
Gradient Descent

$\theta_0 = 0$

$\hat{y} = \theta_1 x$



Cost function



$J(\theta_1) \rightarrow \text{Error} < \begin{matrix} y \\ \hat{y} \end{matrix}$

$\downarrow$

Min.

1. Random θ_1
2. $\hat{y} = \theta_1 x$
3. Error $< \begin{matrix} y \\ \hat{y} \end{matrix}$
4. $\theta_1^{new} = \theta_1^{old} - \alpha \underbrace{\frac{\partial J}{\partial \theta_1}}_{\text{gradient}}$

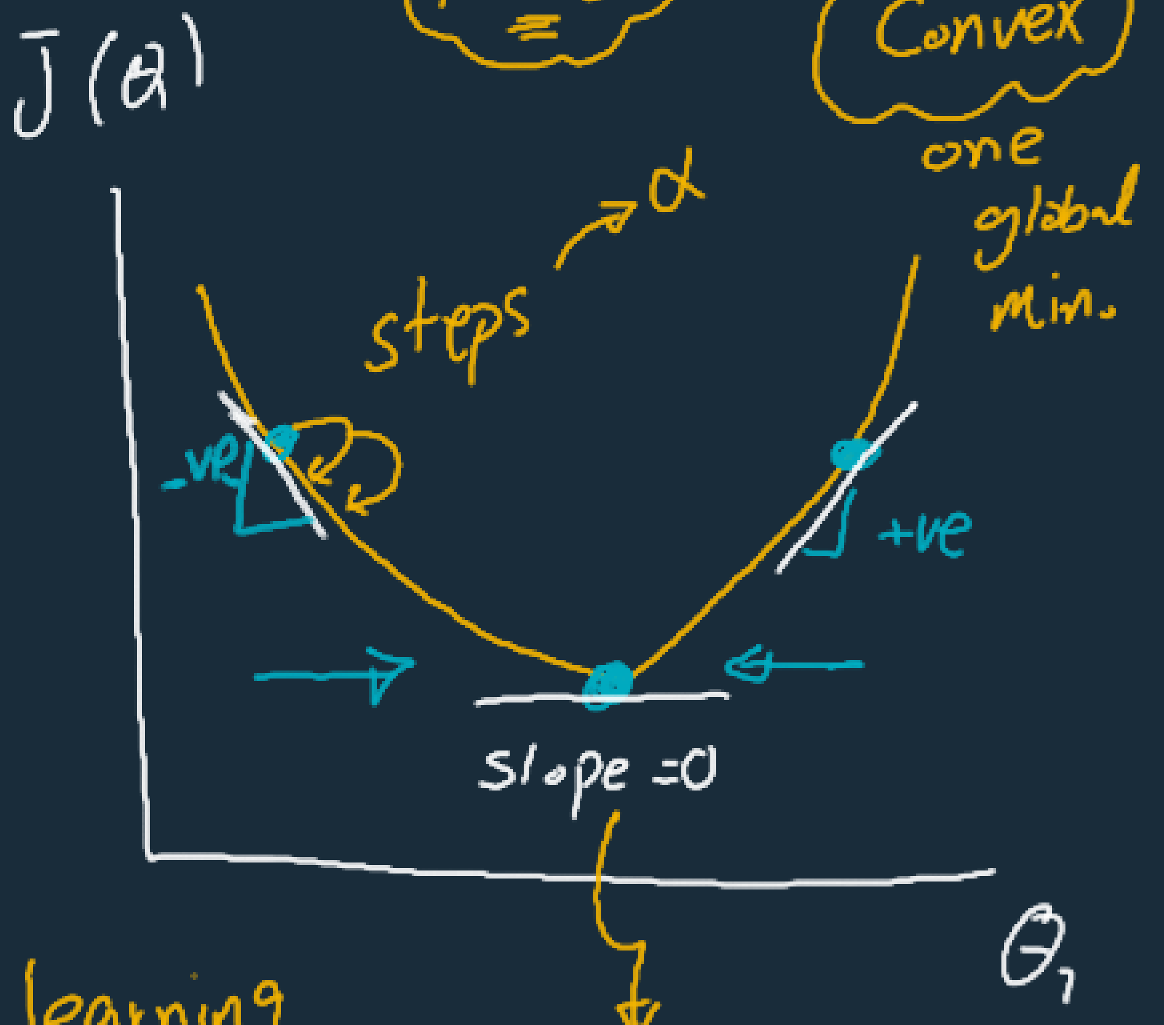
α → learning rate

$\theta^{new} < \theta_{old} \leftarrow +ve$

$\theta^{new} > \theta_{old} \leftarrow -ve$

gradient

MSE



$\theta^{new} = \theta^{old}$

→ Best theta

GD $\rightsquigarrow$ Feature Scaling

?? points each iteration

full data → Batch

Some points → Mini-Batch

one point → Stochastic

θ_1

$X \quad y \quad \hat{y} \quad J(\theta_1)$

Batch grad. descent → slow if large data

Random set → Mini-batch grad. descent

Random point for every iteration → Stochastic grad. descent → fast

Gradient Descent vs. Normal Equation

Gradient Descent	Normal Equation
iterations	No iterations
learning rate	No learning rate
Complexity ↓	Complexity ↑ $\underbrace{(X^T X)^{-1}}$
fast	slow
feature scaling	No need for scaling
SGDRegressor	LinearRegression

No. of features ↑

Sklearn

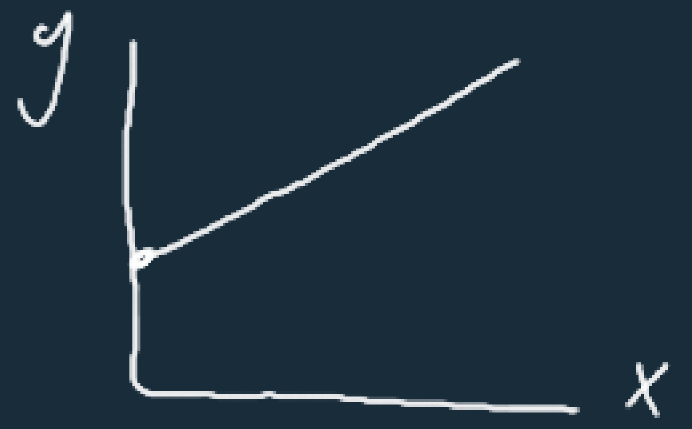
No inverse ? ↓ pseudo inverse

OLS

$y = \theta_0 + \theta_1 X$

↳ one feature

↳ Simple Linear Regression



Normal Eqn

$\hat{y} = \theta^T X$

$$\begin{bmatrix} 1 & \cdots \\ 1 & \cdots \\ 1 & \cdots \end{bmatrix}$$

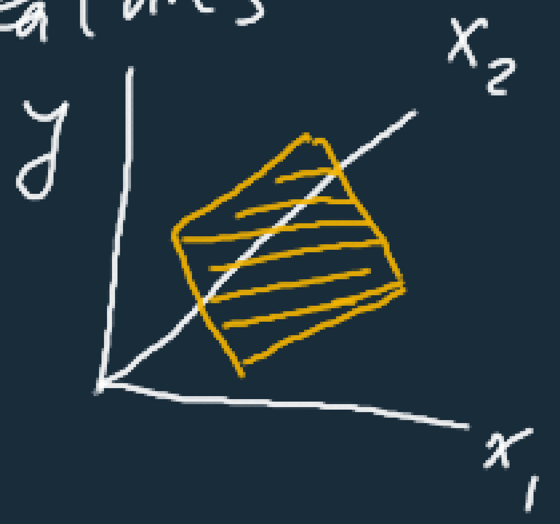
↑

x_0

$$y = \theta_0 \underset{}{\overset{1}{x_0}} + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

weight ← coeff

↳ n features

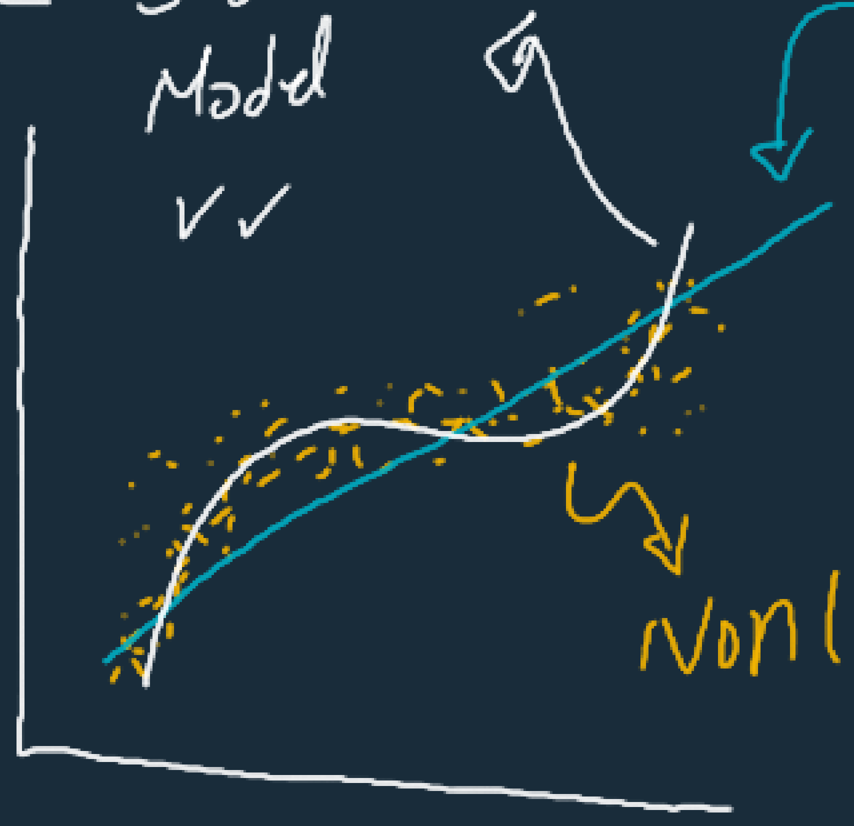


Multiple Linear Regression

Gradient descent → $\dfrac{\partial J}{\partial \theta_0}, \dfrac{\partial J}{\partial \theta_1} \cdots \dfrac{\partial J}{\partial \theta_n}$

$$\theta^{new} = \theta^{old} - \alpha \frac{\partial J}{\partial \theta}$$

Higher Degrees ← Best Model ✓✓



~~Best Model~~

underfitting

↳ Nonlinearity

$$y = \theta_1 x^{2}$$

$$x^{3}$$

$$\theta_1 x^{1} + \theta_2 x^{2}$$

↳ Polynomial Regression

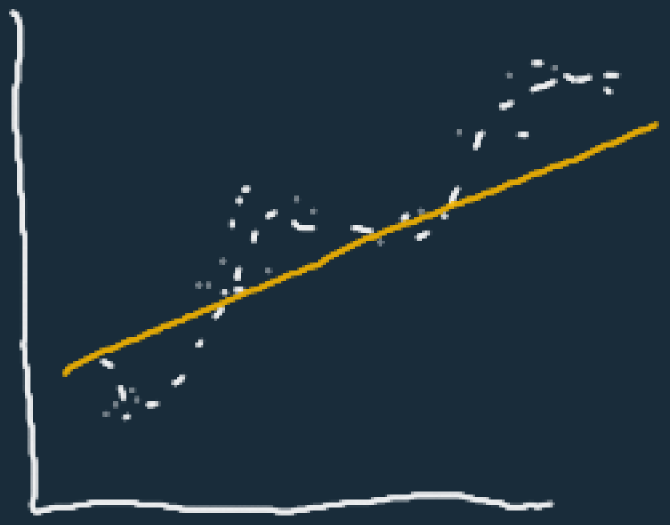
Polynomial Regression

$$\hat{y} = \theta_0 + \theta_1 X_1 + \theta_2 X_1^2 + \theta_3 X_2 X_1 + \theta_4 X_2^2 + \dots \theta_n X_1^d X_2^d$$

Nonlinearity $\left\{ \begin{array}{ll} \text{degree} & 2, 3, \dots \uparrow \\ \text{Combinations} & X_1 X_2 \dots X_1 X_2^2 X_3 \dots \uparrow \end{array} \right\}$ High Complex Model

SKlearn → Polynomial Features (preprocessing) $X \rightarrow \boxed{\text{Poly}} \rightarrow X^d$

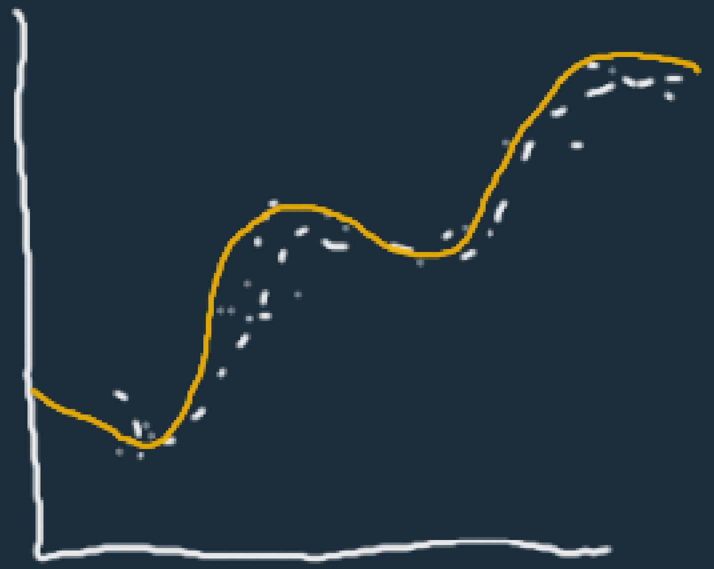
↳ Linear Regression ✓ Polynomial Regression



1

Underfitting

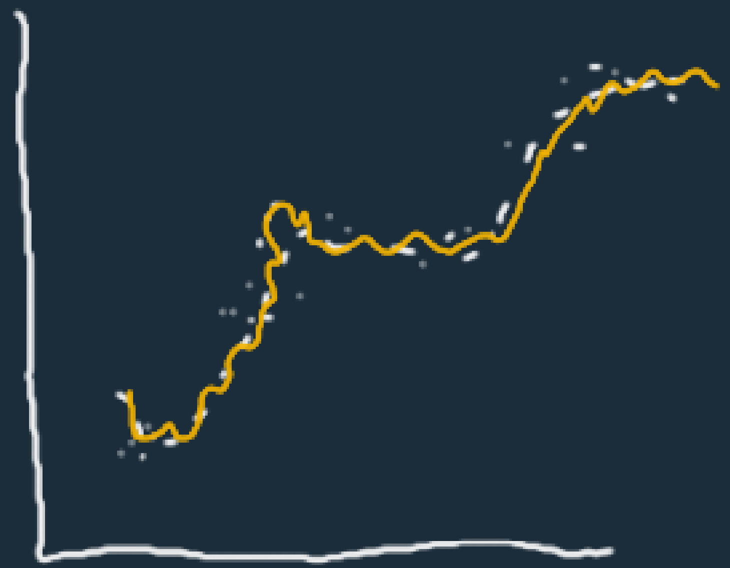
	Error	ACC
Train	↑	↓
Test	↑	↓



2

Good fit

	Error	ACC
Train	↙	↗
Test	↙	↗

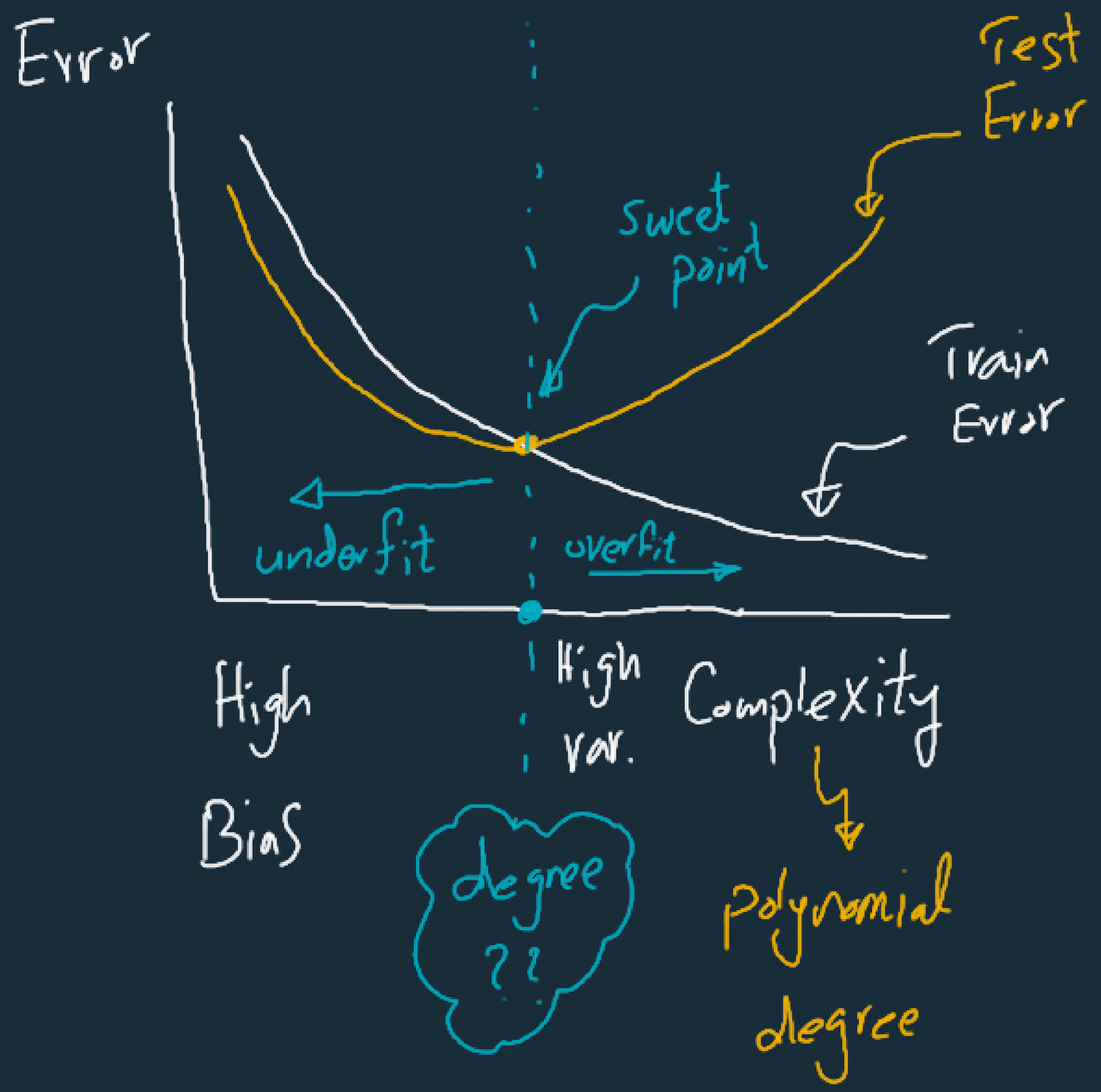


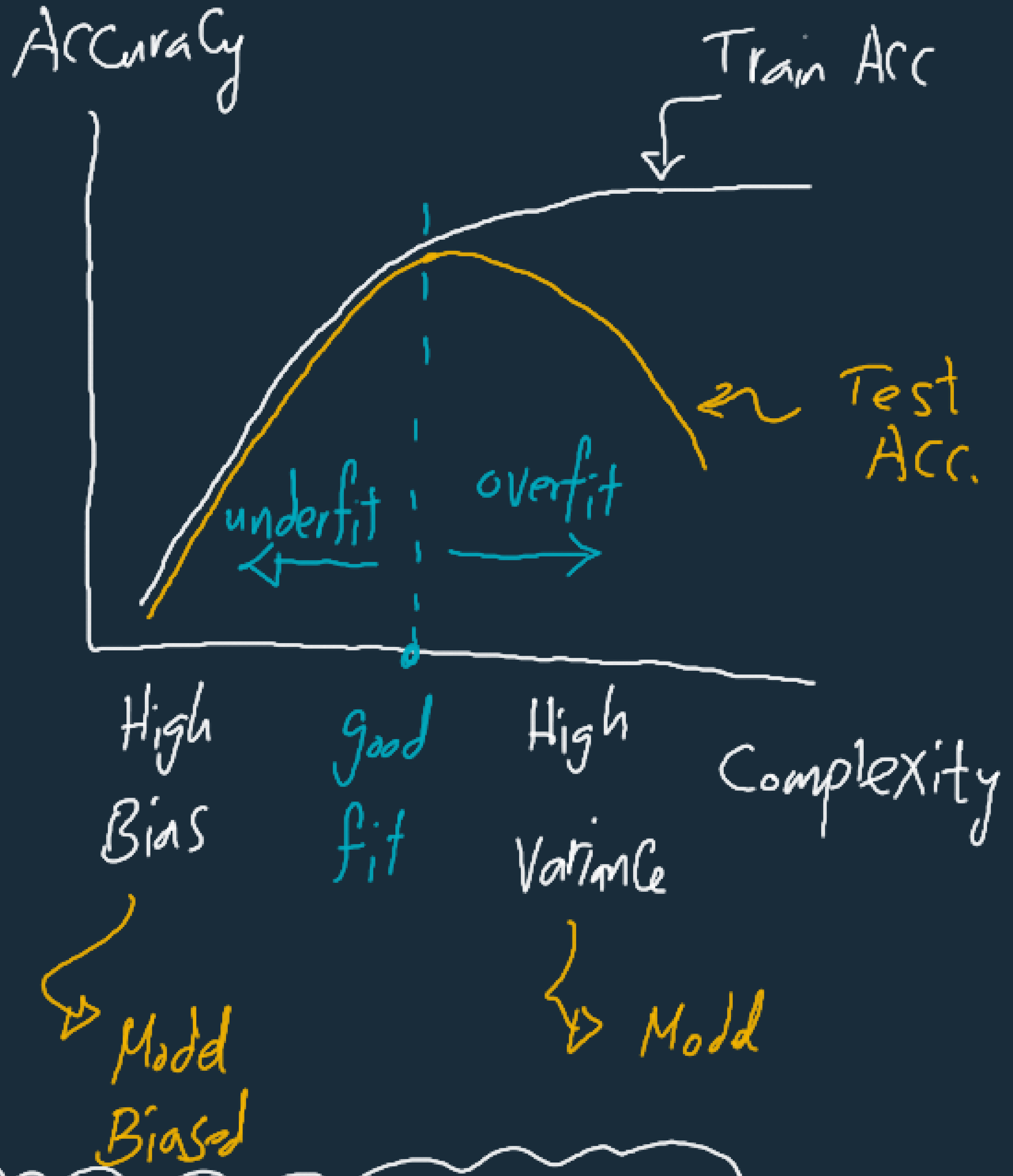
3

overfitting

	Error	ACC
Train	↓	↑
Test	↑	↓

poor generalization





Bias vs. Variance Tradeoff

Degree $\longrightarrow$ Model Complexity

$\downarrow$

Hyperparameter $\rightarrow$ tuning $\rightsquigarrow$ Cross Validation

Grid Search

Complexity $\uparrow\uparrow$ $\longrightarrow$ overfitting $\leftsquigarrow$ Regularization

$\hookrightarrow$ penalty $\longrightarrow$ Regularized Models

Ridge Regression

Regularization $\rightsquigarrow$ Penalty

$$\hat{y} = \theta_0 + \theta_1 X + \theta_2 X^2 + \overbrace{\theta_3 X^3 \dots + \theta_n X^n}^{\text{penalty}}$$

Complexity $\propto$ degree

↳ Higher degrees $\leftarrow$ Penalty

↳ less weight

$\theta_3 = 0.001 \dots 0.0001 \rightarrow X^3 \downarrow\downarrow\downarrow$

$\rightarrow$ Complexity $\downarrow\downarrow$

$\theta \rightarrow$ found during fitting

Gradient Descent

Control Regularization

Cost Function $J(\theta) = MSE + \underbrace{\lambda \sum_{i=1}^{n} \theta_i^2}_{\text{Regularization Term}}$

$\hookrightarrow \lambda = 0 \rightarrow$ No regularization

$\lambda \uparrow\uparrow \rightarrow$ Regularization $\uparrow\uparrow$

$\lambda \uparrow\uparrow\uparrow \rightarrow$ Underfitting

$$\theta_{best} = \begin{matrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \\ \vdots \end{matrix}$$

$\theta_3, \ldots \}$ Higher degrees less weight $\Rightarrow$ ☑ overfitting Solved

Normal Equation

$$\hat{\theta} = (X^T X)^{-1} X^T y \rightarrow \text{No Regularization}$$

Control

$$\hat{\theta} = (X^T X + \underbrace{\lambda A}_{\text{Regularization term}})^{-1} X^T y \rightarrow \text{With Regularization}$$

$A \rightarrow$ Eye matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Gradient Descent

$$J(\theta) = MSE + \lambda \sum_{i=1}^{n} \theta_i^2$$

L2 Regularization (penalty)

Ridge

$$J(\theta) = MSE + \lambda \sum_{i=1}^{n} |\theta_i|$$

L1 Regularization (penalty)

Lasso

→ Ridge $\theta_i \neq 0$

→ Lasso $\theta_i = 0$ (may be) → Feature Selection

$\underbrace{X_1, X_2}_{\text{Feature Selection}}, \overbrace{X_3, X_4}^{\text{No Contribution}}$

$\theta_3 = 0 \quad \theta_4 = 0$

→ Elastic Net

- [x] Linear Regression $\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 \dots$
- [x] Training with GD
- [x] " " OLS / Normal Equation
- [x] Polynomial Features
- [x] Regularized Models
  - Ridge
  - Lasso

overfitting solution